Bayesian Statistics III: Bayesian Linear Regression

ECON 8310: Business Forecasting — Lecture 12

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Lecture Outline

1 Bayesian Linear Regression

2 Reading the Posterior

3 Causal Structure and DAGs

4 Course Synthesis and Next Steps

Bayesian Linear Regression

The same regression you already know — returning a distribution for every coefficient.

OLS vs. Bayesian Regression: What Changes?

OLS, which you have fitted since Lecture 2, solves for one best coefficient vector:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

The Bayesian version uses Lecture 10's update rule instead:

$$p(\boldsymbol{\beta} \mid \mathbf{y}) \;\propto\; p(\mathbf{y} \mid \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta})$$

Every coefficient comes back as a distribution: not an estimate plus an asymptotic error bar, but a full posterior you can ask questions of.

And Lecture 6 reappears: **regularization is a prior**. The posterior *mode* under a Normal prior is the Ridge estimate; under a Laplace prior, it is the LASSO estimate.

Bayesian Linear Regression in PyMC

Log units on log price, SNAP, events, series and trend

```
with pm.Model() as model:
  a       = pm.Normal("a", y.mean(), 1, shape=J)  # one per series
  b_price = pm.Normal("b_price", 0, 1)     # weakly informative
  b_snap  = pm.Normal("b_snap", 0, 1)
  b_event = pm.Normal("b_event", 0, 1)
  b_trend = pm.Normal("b_trend", 0, 1)
  sigma   = pm.Exponential("sigma", 1.0)
  mu = a[idx] + b_price*lpc + b_snap*snap + b_event*event + b_trend*yr
  pm.Normal("obs", mu=mu, sigma=sigma, observed=y)
  idata = pm.sample(1000, tune=1500, chains=4)
```

Do not standardize log price — with both in logs `b_price` is an **elasticity**, percent change in demand per percent change in price. `lpc` is log price centered within series, which fixed ESS 184 $\rightarrow$ 3,097.

Reading the Posterior

Three questions you can ask a distribution that you cannot ask a point estimate.

Reading a Posterior Coefficient

Fitted on the weekly panel, controlling for series and trend:

Coefficient	Mean	94% HDI	Reading
`b_price`	-0.493	$[-0.596, -0.387]$	Inelastic: all of it above -1
`b_trend`	0.110	$[0.107, 0.113]$	About 11% growth per year
`b_snap`	0.036	$[0.027, 0.046]$	About a 3.7% SNAP-week lift
`b_event`	-0.003	$[-0.011, 0.004]$	**Spans zero**

The **HDI** — highest density interval — is the shortest interval holding 94% of the posterior, and it means what everyone wants: a 94% probability the coefficient is inside.

The last row is the useful one. `b_event`'s interval spans zero, so the data do not establish a direction — but it also shows the effect is *small*, between about -1.1% and $+0.4\%$. That is more honest than "not significant."

Asking the Posterior a Direct Question

Any question about β is answered by counting samples.

One line each

```
b = idata.posterior["b_price"].values.ravel()

(b < 0).mean()     # P(demand falls when price rises)
(b < -1).mean()    # P(elastic: a price rise loses revenue)
(b < -0.5).mean()  # P(a 1% rise costs over 0.5% of volume)
```

On our fit, $P(\text{elasticity} < 0) = \mathbf{1.00}$ and $P(\text{elasticity} < -1) = \mathbf{0.00}$: demand falls when price rises, and is **inelastic** in every draw. Whether a 1% rise costs over half a percent of volume stays open: $P(\text{elasticity} < -0.5) = \mathbf{0.46}$.

A p-value reports only the direction. The posterior also answers *how big* — and shows which size questions are still open.

*Posterior mass inside a range of negligible values is a **ROPE** — region of practical equivalence.*

Causal Structure and DAGs

Which coefficients mean what you think they mean.

Directed Acyclic Graphs

A **DAG** is a picture of what you believe causes what: **nodes** are variables, **arrows** run from cause to effect, and nothing causes itself. You draw it before the model, from domain knowledge, to decide *which variables belong in the regression*.

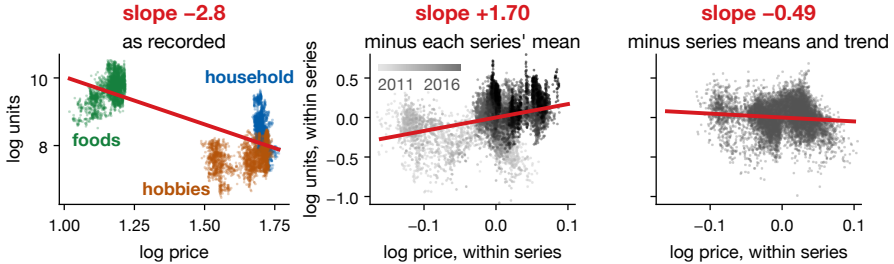
Confounder	Causes both your predictor and your outcome. **Must** be controlled for, or its influence is absorbed into your coefficient.
Collider	Is caused by both. Controlling for it **creates** a spurious association that was not there.

So "add every control you have" is bad advice. Which variables to include is a question about causal structure, and no model fit will answer it.

Whether a DAG is right is a business question, not a statistical one. Estimating causal effects credibly is a separate subject.

Confounding, Measured

Two confounders move price and units together: the **category** and **time**.



Food is cheap and sells in volume; prices and volumes both rose, 2011–2016. Removing the series means alone flips the sign. With both removed: -0.493, HDI $[-0.596, -0.387]$. The naive -2.805, HDI $[-2.856, -2.756]$, is nearly six times too large, at half the width.

> Confidently, precisely wrong: a tight interval says the model is sure, not that it answers your question. Only the DAG protects you from an omitted confounder.

Scenario Analysis with Posterior Samples

Push a scenario through the draws to get a **distribution of outcomes**, not one number.

What if we raise price 5%?

```
b = idata.posterior["b_price"].values.ravel()
pct_change = 100 * (np.exp(b * np.log(1.05)) - 1)
pct_change.mean(), np.percentile(pct_change, [3, 97])
```

With an elasticity near -0.493, a 5% price rise implies roughly a **2.4% fall in units**, 94% of the posterior between -2.9% and -1.9%.

The decision follows. Revenue changes by a factor $1.05^{\,1+b}$, which rises exactly when $b > -1$. Every draw is above -1, so $P(\text{revenue rises}) = 1.00$: about $+2.5\%$ on average.

Valid only to the extent the DAG is — this is a causal claim about intervening on price.

Course Synthesis and Next Steps

From classical time series to Bayesian hierarchical models: a complete forecasting toolkit.

Course Method Map

Method	Data	Uncertainty	Interpretable	Strengths
ETS / ARIMA (L01–02)	Univariate TS	CI	High	Fast, automatic
GAM / Prophet (L03)	TS + predictors	Limited	High	Nonlinear + interpretable
Decision Tree (L04)	Tabular	No	High	Rule-based explanation
Random Forest (L05)	Tabular	OOB error only	Medium	Robust, minimal tuning
XGBoost (L05)	Tabular	No	Low	Interactions; 781 on our board
Ridge / LASSO (L06)	Tabular	No	High	Best on our board: 744
FFN / 1D CNN (L07–08)	Tabular / TS	No	Low	Flexible nonlinear fits
LSTM / Transformer (L09)	Sequences	No	Very low	Long-range dependencies
Bayesian (L10–12)	Any	Full posterior	High	Uncertainty, small samples

No single method dominates. Choose based on: (1) data type and size, (2) need for uncertainty quantification, (3) need for interpretability, (4) time to deploy. In practice, combine: your scoreboard's winner for the point forecast (here, LASSO) and a Bayesian model for the intervals.

Lecture 12: Key Takeaways

1. **Bayesian regression**: a posterior per β, whose mode is Ridge (Normal) or LASSO (Laplace).
2. **Ask the posterior directly**: mean, HDI, $P(\beta_j$ past any threshold$)$.
3. **Draw the DAG first**: control confounders (here, series *and* time), never colliders.
4. **Scenarios**: push a counterfactual through the draws to get a distribution of outcomes.
5. **No single method is best**: pair an accurate point forecast with Bayesian intervals.

References I